

The problem

A tree signals through its leaves months before anyone walking past would notice. In the UAE the pressures are specific: salinity from irrigation water, falling groundwater, Ghaf dieback, red palm weevil, and invasive mesquite displacing the native Ghaf it closely resembles.

Reading those signs is expert work, and there are far more trees than experts.

Our research

We chose ten trees that between them cover most of what grows here, documented the leaf signs for each, and photographed real leaves across Al Ain.

Training images come from openly licensed field photographs, every one credited in the repository. The app was then tested against real leaves rather than only against reference images.

Production stages

Research	Design	Build – v1, Green-Trace UAE	Testing & calibration	Iteration – v2, Warif	Showcase
Term 1, weeks 1-2	Term 1, week 3	2-7 September 2026	7 September 2026	11 September 2026	Term 1 showcase

The pipeline

1 Photograph one leaf on plain paper.	2 Check that there is plant tissue in the frame at all.	3 MobileNetV2 turns the photograph into 1,280 features.	4 Our trained classifier answers with a species and a confidence.
5 An unfamiliarity check can refuse to name it.	6 Separately, classical computer vision grades colour and texture inside the blade.	7 Findings become a likely cause, a treatment, and the nearest shop that sells it.	

Accuracy

Trees recognised	4 / 10
Validation accuracy	81.5%
training images	10,528
held-out validation images	232

Read live from the model's own metadata. The library documents ten trees; the deployed model recognises the number shown, and the rest are marked "Reference only" throughout the app.

Health calibration, v1 to v2

Healthy leaves reported healthy	11.3% – 57.3%
Median score on healthy leaves	47 – 92
Synthetic necrosis detected	99.2%
Synthetic chlorosis detected	89.4%–93.8%

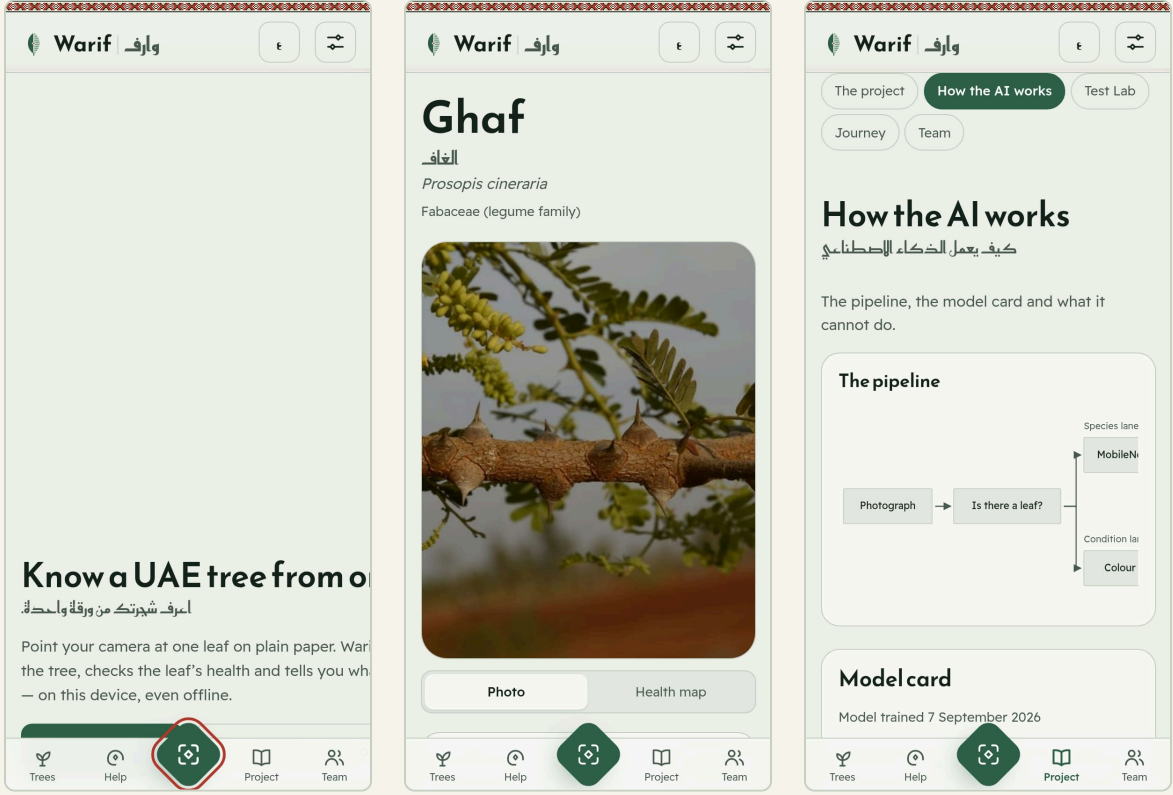
The first health analyser called healthy leaves sick — it was reading the dark background as dead tissue. Measured before and after on the same reference photographs.

Measured on Linux x64 — headless Chromium, SwiftShader, 12 September 2026.

The trees Warif knows

 Ghaf الغاف Recognised	 Sidr السدر Recognised	 Date Palm التاريخ Recognised	 Samar المسمر Recognised	 Grey Mangrove الكرج Reference only	 Athel Tamarisk اثل Reference only	 Arak (Toothbrush Tree) اكرات Reference only	 Neem النيم Reference only	 Apple of Sodom العشور Reference only	 Mesquite المسكيت Reference only
---	---	--	---	---	---	---	---	--	---

The app



Impact

It runs entirely on the device: no account, no upload, no server, and it works with the connection switched off — which is the condition a farm in Al Dhafra or a school in Fujairah is actually in.

Next: finish training all ten trees, add field photographs from Al Ain, and test with a municipality nursery.



Open Warif